

Assignment No. 5

TokenRing.java

```
import java.util.Scanner;

public class TokenRing
{
    public static void main(String args[])
    {
        Scanner sc = new Scanner(System.in);

        int n, token = 0;

        System.out.print("Enter number of processes: ");
        n = sc.nextInt();

        int request[] = new int[n];

        for(int i = 0; i < n; i++)
        {
            System.out.print("Enter request for process " + i + " (0/1): ");
            request[i] = sc.nextInt();
        }

        System.out.println("\nToken Ring Execution:");

        for(int i = 0; i < n; i++)
        {
            System.out.println("\nToken passed to Process " + token);

            if(request[token] == 1)
            {
```

```

        System.out.println("Process " + token + " ENTERS Critical Section");
        System.out.println("Process " + token + " EXITS Critical Section");
    }
    else
    {
        System.out.println("Process " + token + " does not request CS");
    }

    token = (token + 1) % n;
}

    sc.close();
}
}

```

Output

C:\Users\anuja\OneDrive\Documents\DS_prac>cd Assignment5

C:\Users\anuja\OneDrive\Documents\DS_prac\Assignment5>notepad TokenRing.java

C:\Users\anuja\OneDrive\Documents\DS_prac\Assignment5>javac TokenRing.java

C:\Users\anuja\OneDrive\Documents\DS_prac\Assignment5>java TokenRing

Enter number of processes: 4

Enter request for process 0 (0/1): 1

Enter request for process 1 (0/1): 0

Enter request for process 2 (0/1): 1

Enter request for process 3 (0/1): 0

Token Ring Execution:

Token passed to Process 0

Process 0 ENTERS Critical Section

Process 0 EXITS Critical Section

Token passed to Process 1

Process 1 does not request CS

Token passed to Process 2

Process 2 ENTERS Critical Section

Process 2 EXITS Critical Section

Token passed to Process 3

Process 3 does not request CS